

Increasing Chloride Concentration Due to Road Salt In Gwynns Falls Watershed

By: Kai Zimsen-Campe, Jacob Gourley, Scarlet Glass, Ayush Gupta, &
Caroline Averill

Abstract

This project investigates how chloride (Cl) concentrations have changed over time across two different sites in the Chesapeake Bay watershed. These sites include Pond Branch (POBR), a rural forested area used as a reference stream, and Carroll Park (GFCP), an urban watershed in the center of Baltimore. Using long-term monitoring data from the Baltimore Ecosystem Study, weekly samples were taken to evaluate various chemical concentrations. Results show that road salt applications during winter seasons drive spikes in chloride, particularly at the urban site GFCP, where concentrations during winter storms can reach levels toxic to freshwater and the organisms that live in it. These spikes occur primarily during larger winter storm events, when road salt is heavily applied and then quickly washed into the stream system. Overall, the findings of this study demonstrate how impervious surfaces and urban infrastructure amplify chloride pollution, creating a pattern of chemical stress in urban watersheds, while rural systems remain less affected and more resilient.

Introduction

Winter road salt application is a routine practice in urban areas of the US. Its consequences for freshwater systems remain a large concern. Road salt is composed primarily of sodium chloride, which dissolves and releases chloride ions that are chemically conservative, meaning they do not degrade or bind to sediment, allowing them to accumulate in streams and groundwater. In urban watersheds with surfaces such as roads, sidewalks, and parking lots, the salt runoff channels into nearby streams during snowmelt and winter rain. This causes sharp chloride spikes that exceed the EPA's chronic toxicity threshold of 230 mg/L for freshwater life

(EPA). These elevated concentrations create high risks for aquatic organisms while also degrading drinking water quality and corroding infrastructure. In rural forested watersheds, there is a lot less surface runoff, which allows water to pass through the soil and filter contaminants. This results in much lower chloride levels. Over long periods of time, this overuse of salt in urban streams has been linked to Freshwater Salinization Syndrome, a condition defined by rising chloride concentrations even outside of winter (E, K.). The question we looked at was how Chloride concentrations changed over time at POBR (Pond Branch) and GFCP (Carroll Park) within the Chesapeake Bay watershed, and to what extent seasonal spikes and fluctuations reflected road salt usage and runoff.

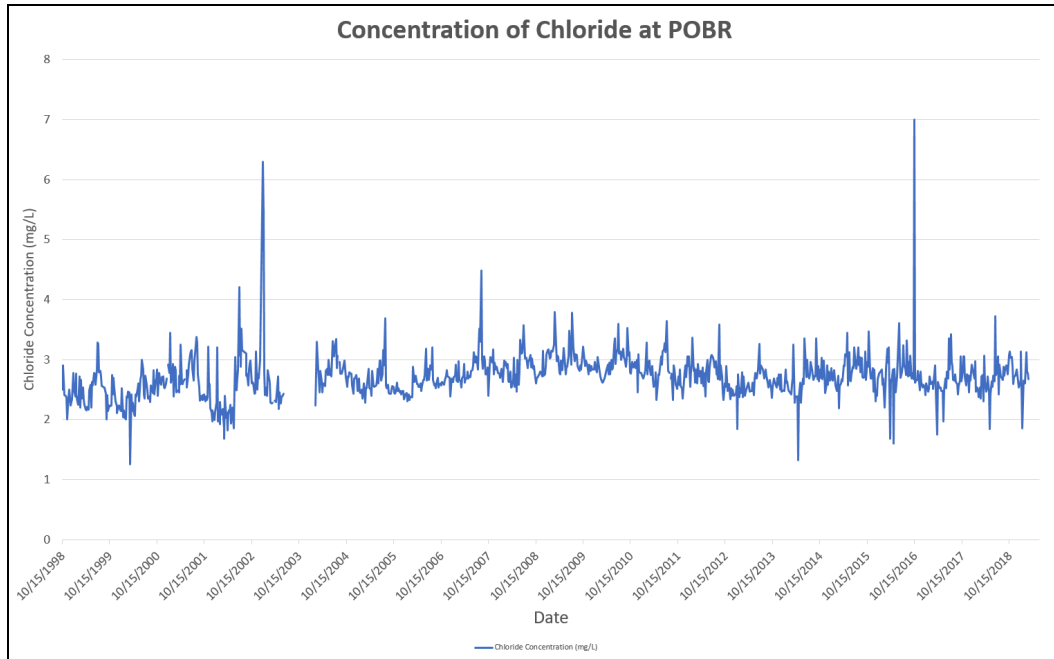
Materials and Methods

Data was taken from *Baltimore Ecosystem Study: Stream chemistry for core sites in Gwynns Falls* (Groffman), which is part of the Baltimore Ecosystem Study (BES) Long Term Ecological Research (LTER) program, and presented by the Environmental Data Initiative (EDI). The study ran from 1988 to 2018, with publication in 2020, in the Gwynns Falls Watershed in Baltimore, Maryland. They used the watershed to evaluate the Baltimore ecosystem. Many stream sites were continuously monitored long-term and sampled weekly or periodically for water chemistry. This analysis focused on the variables date, chloride (Cl), sodium concentration (Na^+), and temperature. The measurements we analyzed were collected by BES researchers through water sampling. Ion concentrations were analyzed in the lab, and temperature was measured in the field when samples. The date was used to analyze long-term trends. The methods followed the BES and LTER protocols. Full methodology is available in the dataset

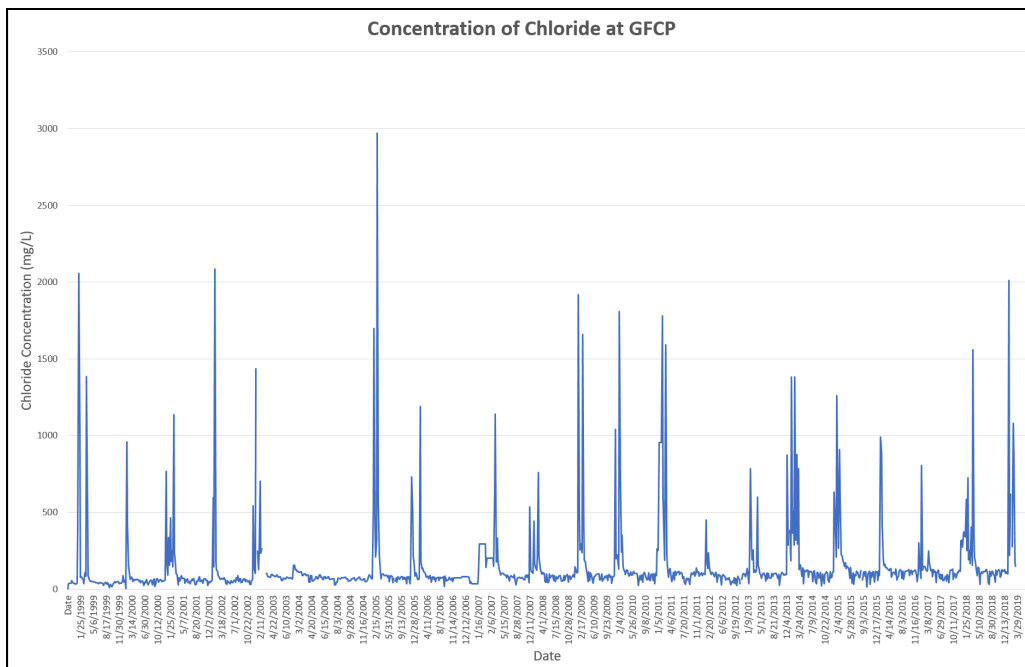
metadata provided through the EDI website under Baltimore Ecosystem Study: Stream chemistry for core sites in Gwynns Falls Metadata.

Results

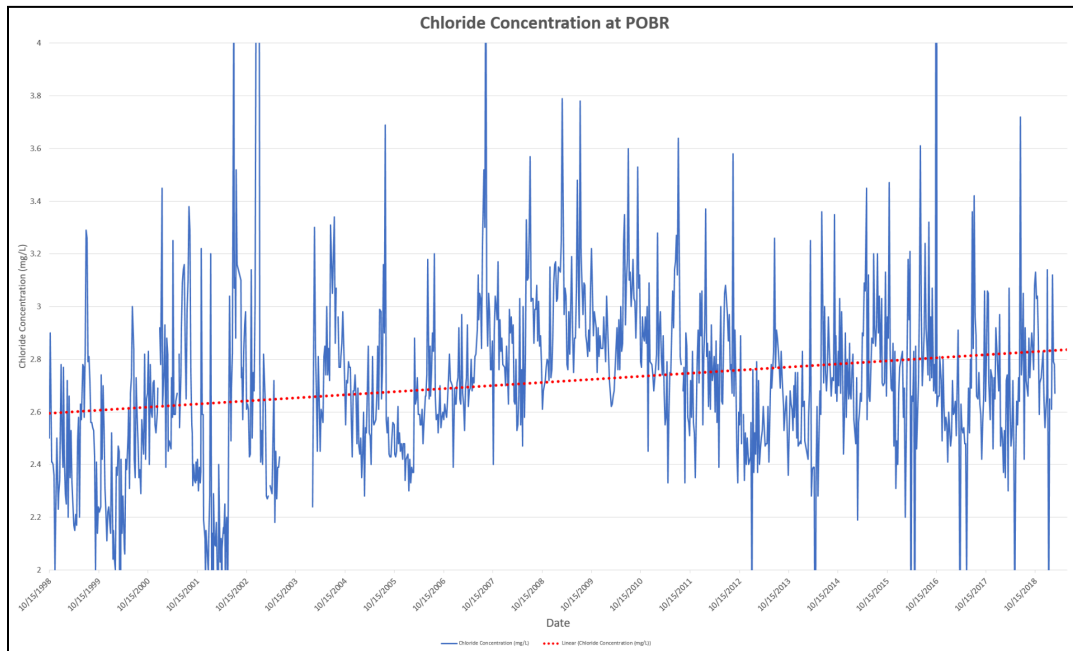
After data was taken from the various sites in the *Baltimore Ecosystem Study*, it was concluded that the concentration of road salt (chloride Cl and sodium concentration Na⁺) has increased over time, with very strong seasonal variability applied to this runoff process. At the Pond Branch (POBR) watershed, there is a low baseline concentration (around 2-3 mg/L). But we found that there are occasional sharp spikes between the early 2000s and mid 2010s, indicating that short-term increase is likely associated with warmer years that caused lots of ice melt. The Carroll Park (GFCP) watershed shows more consistently higher and more frequent chloride concentrations. Because of the GFCP's rural area, there is a higher amount of roadsalt applied and more surfaces where runoff can occur, leading to higher peak concentrations during winter events. During winter storms, chloride levels rise and, depending on the environment, flow into different bodies of water once it is melted. This harms the ecosystems and organisms living in this fresh water. This has a very negative impact on certain environmental aspects, and it will continue to harm the organisms.



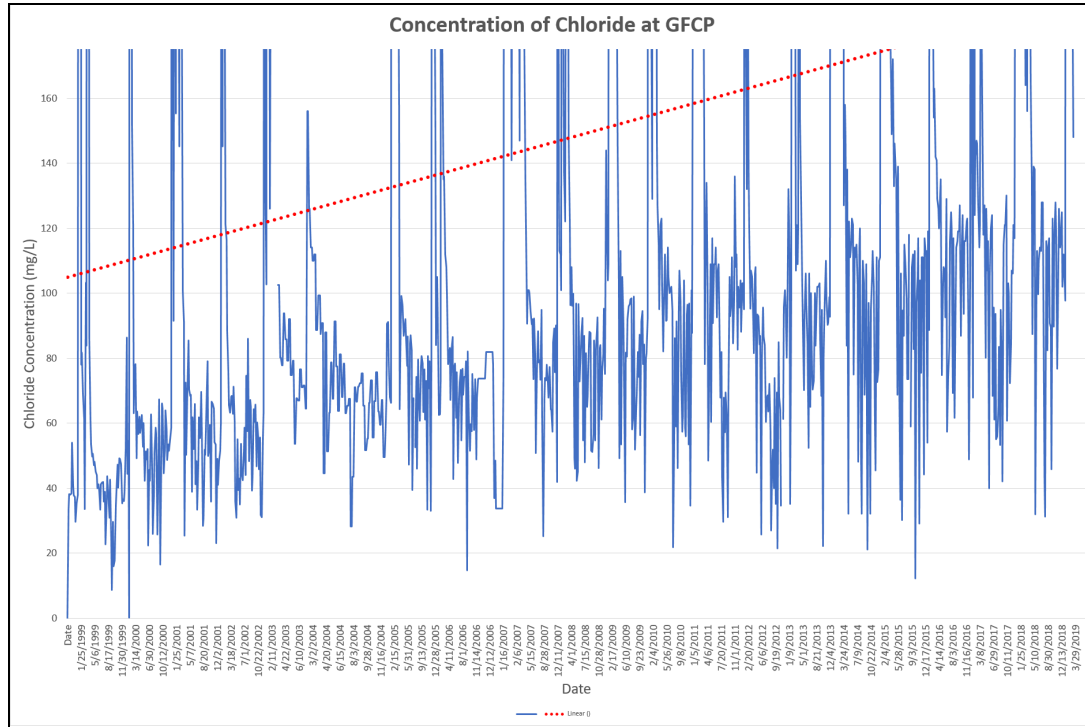
The Chloride concentration at POBR is consistently low, with two spikes occurring at the beginning and end. The spikes indicate major snowstorms that could have occurred, which would cause more salt to be released.



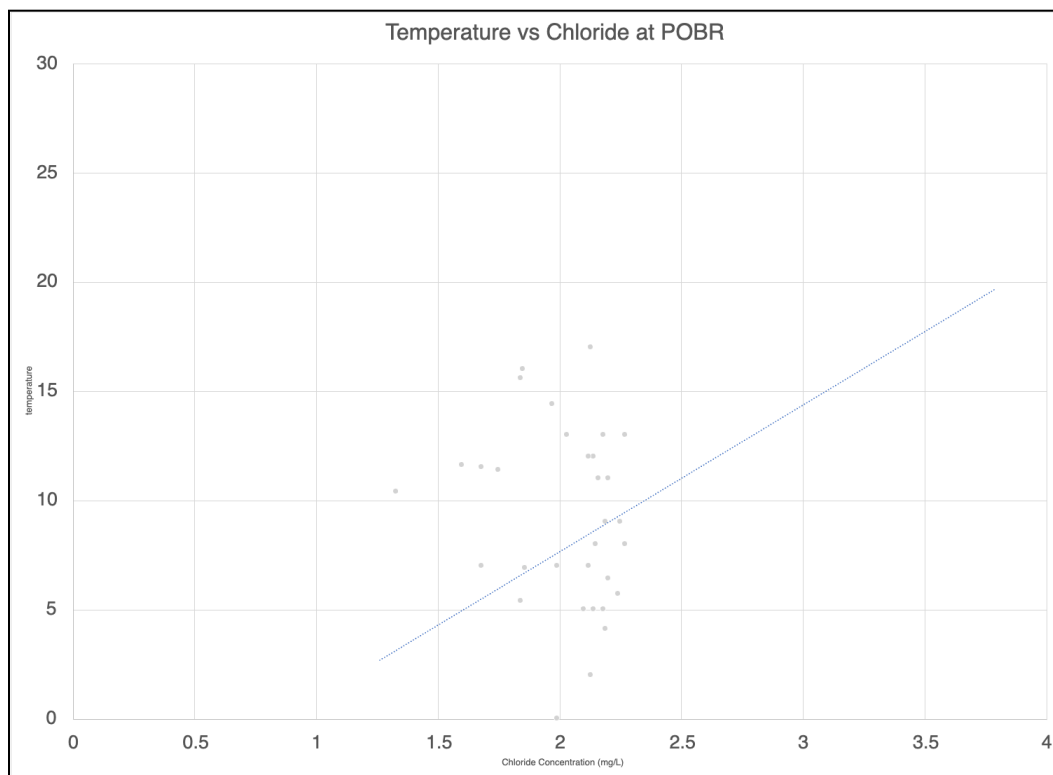
The Chloride concentration at GFCP, in contrast to the graph above, shows a lot more spikes in the graph, which tells us that the area where this occurred had more winter storms and therefore more salt on the roads.



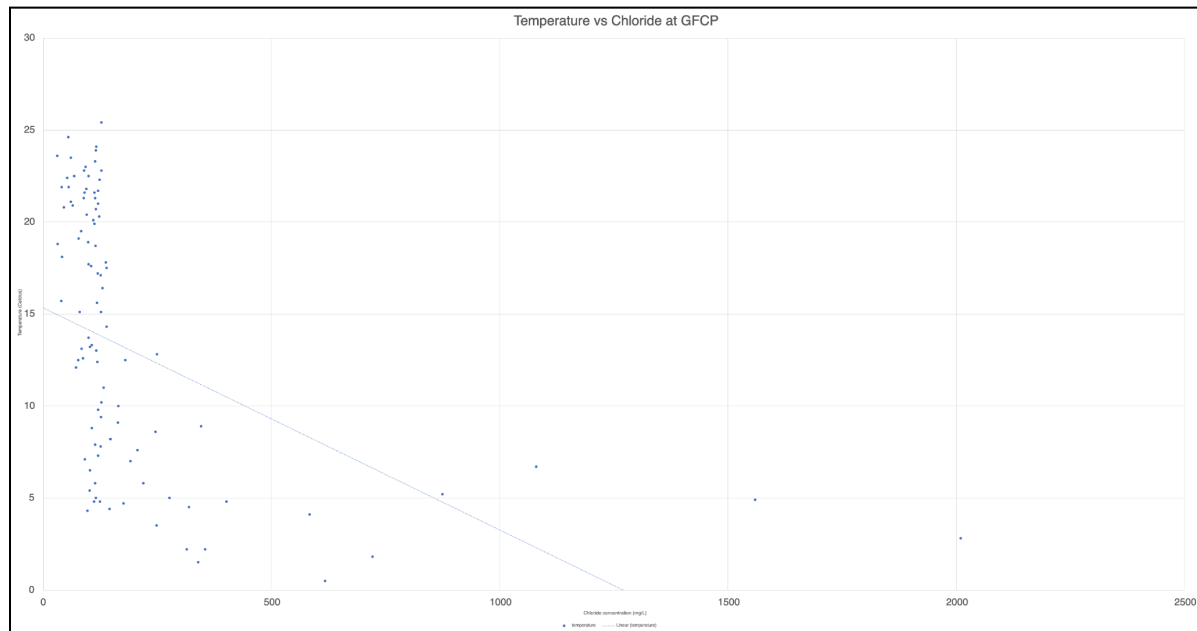
The graph shows a slight, steady increase over time at POBR. This baseline can be compared to GFCP, which trends at much higher values.



This graph shows a more aggressive increase in chloride compared to the previous graph for POBR.



This graph compares temperature and chloride. The positive slope indicators when the temperature increases, the chloride also increases (a clear positive linear relationship between the two variables).



This graph compares temperature and chloride. The negative slope indicates that when temperature decreases, chloride concentration increases.

Discussion

Looking at the chloride concentration over time graphs for both sites, POBR shows a consistently low and relatively flat pattern across the full 1998-2018 study period, with only occasional sharp spikes. GFCP, by contrast, shows much higher and more frequent spikes throughout the same time period, with concentrations regularly reaching into the hundreds and even thousands of mg/L during winter months. This difference in the overall shape of the data between the two sites already suggests that urbanization plays a major role in chloride loading. This makes sense given that GFCP drains a watershed of roughly 66 square miles with about

40% impervious surface cover and a dense road network, while POBR drains only about 0.15 square miles of mostly forested land with very little impervious surface (Groffman).

When the spike values are removed, and only the baseline concentrations are compared, the two sites are not comparable to each other. POBR maintains a stable baseline of around 2-3 mg/L throughout the study period, while GFCP's baseline sits consistently and significantly higher, even during warmer months when no road salt is being applied. This suggests that GFCP experiences chronic salt loading beyond just storm events, likely due to chloride that has built up in groundwater and soil over many years of repeated salt application. POBR's low and stable baseline confirms its role as a reference stream, and shows that the elevated levels at GFCP are driven by urbanization rather than natural conditions. In fact, comparing the beginning of the dataset to the end, POBR shows remarkable resilience: water quality in 2017 is chemically very similar to what it was in 1998.

The temperature vs. chloride scatter plots further support this pattern. At GFCP, the trendline shows a negative correlation, meaning that as temperatures drop, chloride concentrations increase. The highest concentrations at GFCP occur when water temperatures are at their coldest, around 0.3°C, which confirms that the chloride levels are driven by human activity rather than natural seasonal processes like evaporation or mineral weathering. At POBR, this relationship is much weaker, which reflects how the surrounding forested land is able to filter runoff through the soil before it reaches the stream.

Several of the largest chloride spikes in the data can be matched to specific winter storm events. At GFCP, a spike to 2,056 mg/L on January 9, 1999, lines up with a major winter storm that hit the Baltimore area that month (Smith 1-6). The spike to 2,084 mg/L on January 22, 2002,

matches a significant storm recorded in National Weather Service storm data from that time (National Weather Service 1-5). The largest recorded spike in the entire dataset, reaching nearly 3,000 mg/L around February 2005, and a spike to 2,010 mg/L on January 14, 2019, corresponds to a widespread ice and freezing rain event that affected much of the eastern United States (CNBC). All of these spikes are well above the EPA's chronic toxicity threshold of 230 mg/L for freshwater life, meaning these events likely caused serious harm to aquatic organisms at GFCP (EPA). It is also important to note that these repeated chemical shocks may be more damaging to freshwater organisms than constant elevated exposure, since organisms are not given enough time to recover between events. At POBR, smaller spikes to 6.3 mg/L on January 7, 2003, and 5.49 mg/L on January 14, 2003, line up with the start of Baltimore's severe winter of 2003, which included the President's Day Blizzard in February (National Weather Service 1-5). While these spikes are much smaller than those at GFCP, they show that even forested reference streams can be affected during extreme winter events.

One of the most concerning findings is the overall upward trend in chloride concentrations at GFCP over the full study. The best fit line in the zoomed concentration graph shows a slow but steady increase in chloride levels from the late 1990s through 2018, even outside of the seasonal spikes. This is consistent with Freshwater Salinization Syndrome, where repeated yearly salt applications build up in groundwater and soil over time, slowly raising background chloride levels even in warmer months when no road salt is being used (E,K.). If this trend continues, GFCP could face elevated chloride levels that stress aquatic life throughout the entire year, not just during winter storms. POBR's best fit line stays mostly flat over the same period, which shows how forested watersheds are more resilient and less impacted by long term salt accumulation.

Overall, these findings show that the amount of impervious surface, storm intensity, and how much salt is applied over time all contribute to chloride levels in urban streams. Reducing road salt use, or adding more permeable surfaces, could help lower these impacts, especially in urban watersheds like GFCP where chloride levels regularly exceed federal toxicity thresholds during the winter.

References

- E, K. (Ed.). (2022, November 29). *EPA researching the impacts of freshwater salinization syndrome*. Epa.gov. Retrieved May 4, 2026, from <https://www.epa.gov/sciencematters/epa-researching-impacts-freshwater-salinization-syndrome>
- Groffman, P.M., E. Rosi, and L.D. Martel. 2020. Baltimore Ecosystem Study: Stream chemistry for core sites in Gwynns Falls ver 601. Environmental Data Initiative. <https://doi.org/10.6073/pasta/45129da171f2a8ab5a96e9743d0d644b> (Accessed 2026-05-03).
- National recommended water quality criteria - aquatic life criteria table*. (2026, February 13). United States Environmental Protection Agency. Retrieved May 4, 2026, from <https://www.epa.gov/wqc/national-recommended-water-quality-criteria-aquatic-life-criteria-table>
- National weather service*. (2002, January). Weather.gov. Retrieved May 4, 2026, from <https://www.weather.gov/media/lwx/stormdata/2002/storm0102.pdf>
- National weather service*. (2003, January). Weather.gov. Retrieved May 4, 2026, from <https://www.weather.gov/media/lwx/stormdata/2003/storm0103.pdf>
- Pavloff, D. (2005, December 10). *To Salt or Not To Salt? Roadways & Parking Lots Threaten Freshwater*. Cary Institute of Ecosystem Studies. Retrieved May 4, 2026, from <https://www.caryinstitute.org/news-insights/feature/salt-or-not-salt-roadways-parking-lots-threaten-freshwater>

Smith, C. (Ed.). (1999, January 18). *US it's on-again, off-again power for washington area*.

Cnn.com. Retrieved May 4, 2026, from

<https://www.cnn.com/WEATHER/9901/18/dc.weather/index.html>

Storm punishes swath of US with snow, ice and freezing rain. (2019, January 14). Cnbc.com.

Retrieved May 4, 2026, from

<https://www.cnn.com/2019/01/14/storm-punishes-swath-of-us-with-snow-ice-and-freezing-rain-.html>

Tasselmyer, T. (Ed.). (2019, January 14). *Cleanup continues, schools closed after weekend snow*.

Wbaltv.com. Retrieved May 4, 2026, from

<https://www.wbaltv.com/article/tom-previews-weekend-snow-forecast/25838870>